

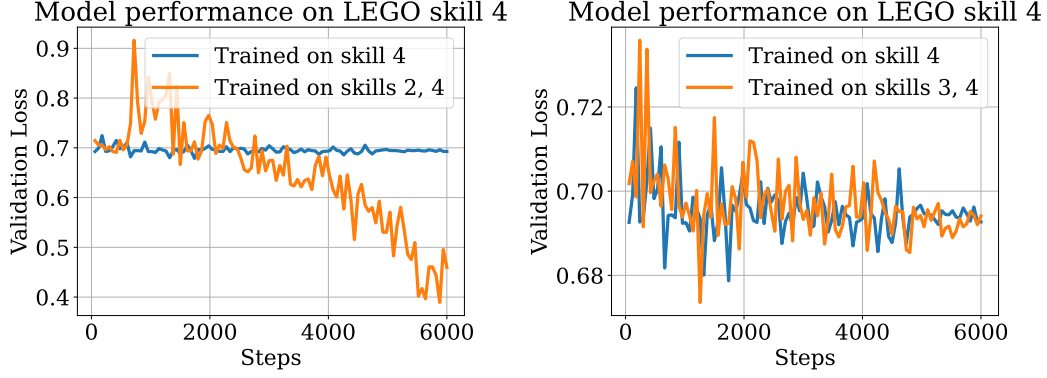


Figure 16: Performance on LEGO skill 4 when training on skill 4, skills 2 and 4, and skills 3 and 4. Even though skill 3 and skill 4 share an edge in the LEGO synthetic’s underlying reasoning chain (i.e. a model predicting correct for the fourth variable is one extra step beyond predicting correct for the third variable), we find that training on skills 2 and 4 helps improve performance on skill 4 more.

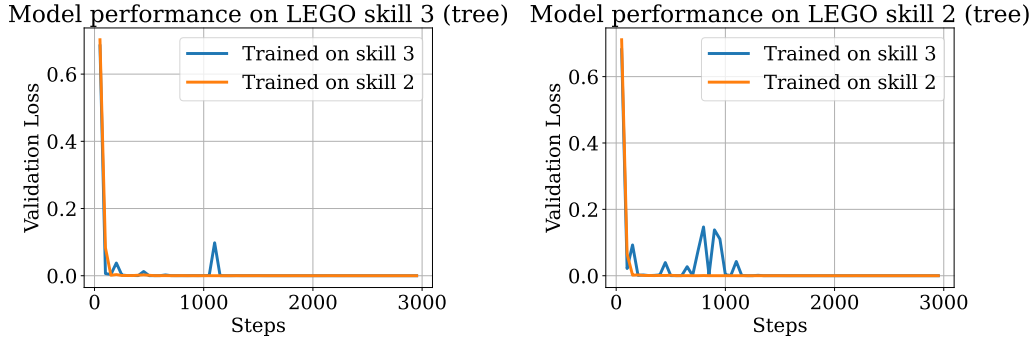


Figure 17: Performance on LEGO skill 2 and 3 when training on skills 2 and 3. The reasoning pattern is a tree rather than a chain over $k = 4$ variables. Skills 2 and 3 are at the same “depth” in the graph and both depend on skill 1, so there is positive influence between the skills despite there being no edge between 2 and 3 in the LEGO reasoning graph.

D Additional Experimental Results

D.1 Additional examples of LEGO ordered skill sets

For the LEGO synthetic, it may appear obvious that the skills graph is equivalent to the reasoning chain over the variables. However, in Figure 16 we see that this is not the case. Training on skills 2 and 4 together results in lower loss on skill 4 than when trained on skill 4 alone. However, training on skills 3 and 4 together results in roughly the same loss on skill 4 as when training on skill 4 alone, even though skill 3 and skill 4 share an edge in the LEGO synthetic’s underlying reasoning chain. This suggests that our intuition for how skills influence each other does not always match how the model learns skills.

Next, we consider a slightly more complex reasoning pattern on the LEGO synthetic. Instead of a chain, we construct a tree, where two variables in the LEGO synthetic are both defined in terms of the same parent variable. For example,

Input: $c = \text{val } 1$, $y = \text{not } w$, $v = \text{val } c$, $w = \text{not } c$. Output: $y = 1$.

In this example, $k = 4$ and both v and w are written in terms of c , and the reasoning graph has edges $1 \rightarrow 2$, $1 \rightarrow 3$, $2 \rightarrow 4$. In this case, we see that training on skill 2 or skill 3 both improve losses on skills 2 and 3 (Figure 17). However, unlike the previous figures, training on skills 2 and 4 or skills 3 and 4 do not significantly help reduce loss on skill 4 (Figure 18). Again, these measurements demonstrate that the reasoning graph does not necessarily equal the skills graph.

D.2 Unsupervised skill recovery

We explore several clustering techniques for recovering the skills in the LEGO synthetic on the validation dataset. Our results are shown in Table 3.

We first cluster based on the pre-trained model embeddings of the last token and the average token. We also report

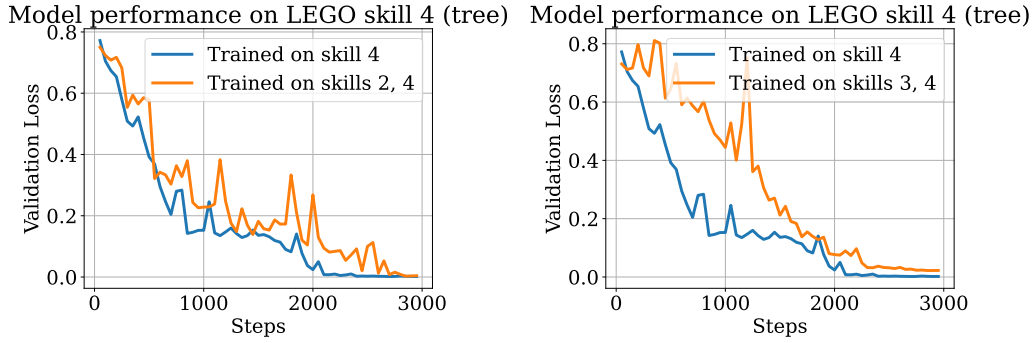


Figure 18: Performance on LEGO skill 4 when training on skills 2, 4 and skills 3, 4. We find that in both cases, the benefit from training on additional skills is minor. For instance, training on 2 and 4 reaches 0.01 loss in 2700 steps, while training on 4 only reaches it in 2100 steps.

Cluster method	Accuracy
Pretrained embedding of last token	24.8 ± 0.5
Pretrained embedding of average token	25.2 ± 1.1
Trained model embedding of last token	38.4 ± 0.8
Sentence-BERT embedding	23.9 ± 0.7
Losses over multiple runs	61.0 ± 1.6

Table 3: Clustering-based skill recovery methods on the LEGO dataset. The validation dataset we cluster consists of 500 points with $k = 5$, and results are reported over 10 runs of k-means.

accuracies of clustering based on the trained model embedding’s last token, where we train the model using random sampling for 6000 steps, and clustering based on Sentence-BERT embeddings. Among these four methods, using the trained model embeddings has the highest accuracy of 38.4 points.

Next, we cluster points based on losses. In particular, we do 10 runs, each for 6000 steps and with a randomly sampled mixture of skills. For each run, we evaluate the model on the validation dataset at 120 checkpoints. Then, each sample in the validation dataset has 1200 losses associated with it, comprising a feature vector for that sample. We perform k-means clustering on these features, which has an accuracy of 61.0 points, significantly higher than the second best accuracy of 38.4.

D.3 Full results for Section 4

D.3.1 Per-skill performance

In this section, we provide tables containing the per skill break-down of our results from Section 4.

Continual Pre-training In the continual pre-training setting, we report two additional baselines that combine curriculum learning with skills. Curriculum learning has been proposed for multitask learning [60], in which groups of data are ranked by their average score and then trained in order of this ranking (with mixing of previously seen groups to avoid forgetting). We construct two baselines, Skill-curriculum and Skill-anticurriculum, using Algorithm 1 from [60]. In contrast to the random baseline which has imbalanced skills, this approach has knowledge of skills and thus uses a skill-stratified training dataset to sample from. We set the fraction of the previous group to be $\text{frac} = 0.4$, as we found that setting $\text{frac} = 0.0$ resulted in forgetting.

We report loss per skill for the LEGO synthetic in Table 4, which corresponds to the results in Figure 4. We report accuracy per skill in Table 5 and Figure 19. We report the loss per skill for the Addition synthetic in Table 6, which also correspond to the results in Figure 4. Finally, we report validation loss per task category for the Natural Instructions continual pre-training experiment in Table 7, where we find that SKILL-IT outperforms random sampling by 3.2% on average across skills.

	Skill 1	Skill 2	Skill 3	Skill 4	Skill 5	Average
Random	0 ± 0.000	0.675 ± 0.041	0.688 ± 0.008	0.673 ± 0.049	0.667 ± 0.056	0.541 ± 0.031
Curriculum	0 ± 0.000	0.645 ± 0.052	0.686 ± 0.018	0.674 ± 0.042	0.671 ± 0.0459	0.535 ± 0.029
Anticurriculum	0 ± 0.000	0.690 ± 0.003	0.695 ± 0.004	0.693 ± 0.003	0.689 ± 0.004	0.554 ± 0.001
Skill-stratified	0 ± 0.000	0.045 ± 0.036	0.056 ± 0.029	0.079 ± 0.044	0.050 ± 0.025	0.046 ± 0.022
Skill-curriculum	0 ± 0.000	0.484 ± 0.200	0.698 ± 0.027	0.697 ± 0.010	0.689 ± 0.007	0.514 ± 0.040
Skill-anticurriculum	0.001 ± 0.001	0.174 ± 0.118	0.245 ± 0.091	0.443 ± 0.125	0.566 ± 0.118	0.286 ± 0.060
SKILL-IT	0 ± 0.000	0.002 ± 0.002	0.024 ± 0.031	0.013 ± 0.010	0.022 ± 0.021	0.012 ± 0.008

Table 4: Results on validation loss per skill for LEGO pre-training experiment, averaged over 5 random seeds.

	Skill 1	Skill 2	Skill 3	Skill 4	Skill 5	Average
Random	100.0 ± 0.0	54.2 ± 5.9	58.0 ± 3.1	48.0 ± 6.3	54.4 ± 7.3	62.9 ± 3.5
Curriculum	100.0 ± 0.0	60.0 ± 10.6	55.2 ± 5.8	51.2 ± 6.3	51.8 ± 6.1	63.6 ± 3.6
Anticurriculum	100.0 ± 0.0	53.4 ± 2.3	49.0 ± 4.8	48.2 ± 6.4	56.0 ± 5.7	61.3 ± 2.2
Skill-stratified	100.0 ± 0.0	98.2 ± 1.8	98.2 ± 1.3	97.8 ± 1.6	98.2 ± 1.3	98.5 ± 0.9
Skill-curriculum	100.0 ± 0.0	75.2 ± 30.1	52.2 ± 3.7	51.0 ± 4.6	54.4 ± 3.1	66.6 ± 7.7
Skill-anticurriculum	100.0 ± 0.0	90.2 ± 8.1	88.2 ± 8.3	73.2 ± 12.2	62.4 ± 9.4	82.8 ± 4.9
SKILL-IT	100.0 ± 0.0	99.2 ± 0.8	99.0 ± 1.0	99.4 ± 0.5	99.6 ± 0.5	99.4 ± 0.2

Table 5: Results on accuracy per skill (binary classification) for LEGO pre-training experiment, averaged over 5 random seeds.

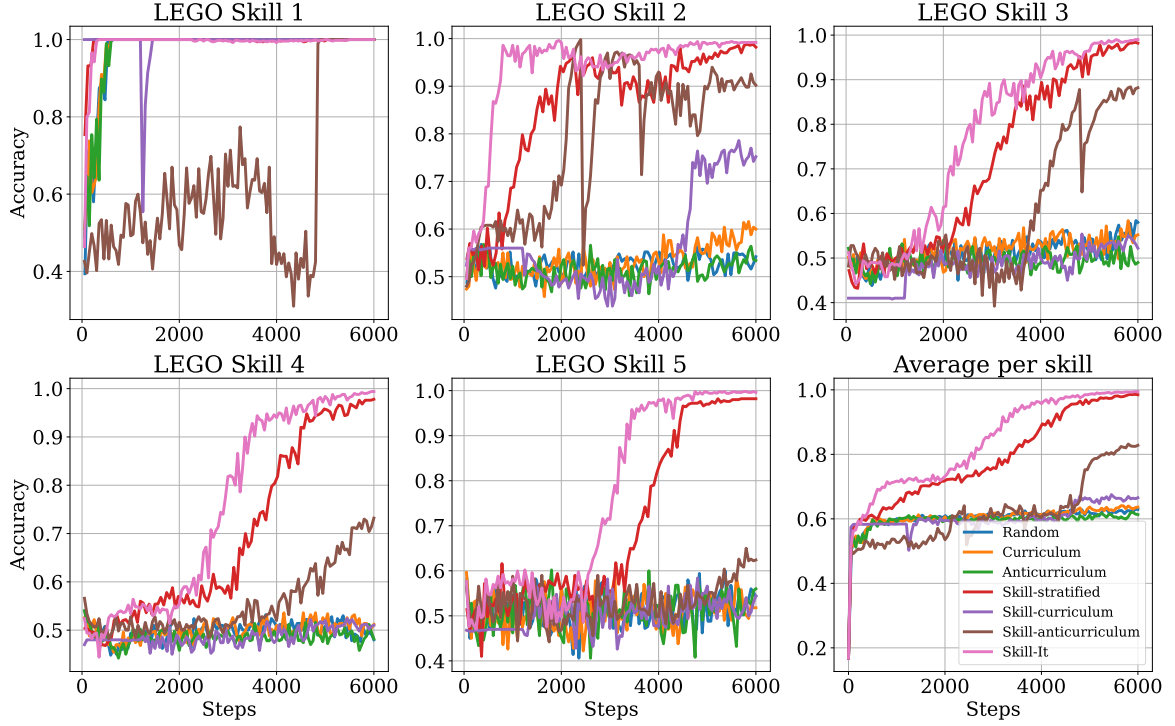


Figure 19: Accuracy of SKILL-IT on each skill (binary classification) on the LEGO synthetic in the continual pre-training setting. SKILL-IT attains higher accuracy more quickly than baselines that both do and do not utilize the notion of skills.

	Skill 1	Skill 2	Skill 3	Average
Random	0.008 ± 0.007	0.020 ± 0.019	0.005 ± 0.005	0.011 ± 0.014
Curriculum	0.009 ± 0.011	0.010 ± 0.008	0.008 ± 0.010	0.009 ± 0.010
Anticurriculum	0.007 ± 0.010	0.012 ± 0.013	0.008 ± 0.017	0.009 ± 0.014
Skill-stratified	0.012 ± 0.011	0.015 ± 0.015	0.010 ± 0.020	0.012 ± 0.016
Skill-curriculum	0.016 ± 0.013	0.019 ± 0.013	0.010 ± 0.003	0.015 ± 0.010
Skill-anticurriculum	0.005 ± 0.008	0.037 ± 0.028	1.141 ± 1.126	0.395 ± 0.371
SKILL-IT	0.004 ± 0.003	0.009 ± 0.007	0.013 ± 0.017	0.009 ± 0.011

Table 6: Results on validation loss per skill for Addition pre-training experiment, averaged over 5 random seeds.